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THEME • DIGITAL TRANSFORMATION IN LENDING

# AI-Powered Hyper-Personalized Banking for Bharat

A point of view on solving India's personalization gap in digital lending — architecture, AI/ML approach, prototype plan and ethical safeguards.

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**How to read this document** — Section 1 restates the brief exactly as given so scope stays anchored to it. Sections 2 onward are our own point of view: how we read the problem, the architecture and models we'd build, and a handful of ideas we're adding on top of the brief because we think they genuinely move the "personalization vs. upsell" needle the judges are looking for.

# 1. The Problem Statement, As Given

## Theme: Digital Transformation in Lending

Indian banks have made massive strides in digital infrastructure — UPI, net banking, mobile apps, video KYC — yet a large section of customers, especially in Tier 2/3/4 towns and rural India, still find banking apps confusing, generic, and disconnected from their actual financial needs. At the same time, banks struggle with rising customer acquisition costs, high drop-offs in loan/product journeys, and difficulty cross-selling the right product to the right customer at the right time. Much of this stems from treating every customer to the same one-size-fits-all digital experience, despite sitting on rich transactional and behavioural data.

## The challenge asks for an AI-powered solution that can:

- **Recommend the right product, at the right moment** — analyse transaction history, spending patterns and life-stage signals (salary credits, EMI patterns, savings behaviour) to proactively surface the most relevant loan, insurance, investment or credit card offer, instead of generic pop-ups.
- **Simplify the journey in vernacular languages** — use conversational AI/chatbots to personalise onboarding, loan application and KYC, cutting drop-offs for first-time digital users.
- **Catch financial stress or fraud early** — detect unusual transaction patterns, missed EMIs or sudden behaviour change, and respond with proactive, empathetic interventions rather than punitive default flags.

## Deliverables expected

- A conceptual architecture / flow diagram: data sources → AI engine → personalized output/action.
- A working prototype or wireframe of the customer journey (dashboard, chatbot flow, or loan journey).
- An explanation of the AI/ML approach (recommendation engines, behavioural segmentation, anomaly detection, vernacular NLP).
- A note on ethical safeguards — data privacy and consent (DPDP Act, RBI data localization), algorithmic bias, and avoiding predatory nudging.

## Judging criteria

- Innovation and technical feasibility
- Depth of personalization vs. genuine customer benefit (not just upsell-driven)
- Explainability and RBI/regulatory compliance readiness
- Usability for non-tech-savvy and vernacular-first users
- Scalability across a bank's existing digital infrastructure, and social/financial impact

## 2. Our Point of View on the Problem

We don't read this as an “AI feature” problem — recommendation engines and chatbots already exist in most bank apps. We read it as a **trust and relevance** problem. A sugarcane farmer in Maharashtra and a software engineer in Bangalore are shown the same dashboard, the same English-first copy, and the same generic “get a personal loan” banner — regardless of whether either of them actually needs it right now. The bank has the data to know better; it just isn't using it to serve the customer, only to sell to them.

So the design principle we're anchoring everything to is: **every AI-driven action must pass a “would we show this to our own grandmother in her language, and would it help her more than it helps the bank?” test.** Concretely, that means three things run through every module below:

- **Moments over pop-ups** — recommendations are tied to a detected life event or cash-flow moment (salary day, EMI ending, surplus balance), not a sales calendar.
- **Vernacular is the default, not an add-on** — the chatbot and every alert are designed language-first, not translated as an afterthought.
- **Stress silences the sales engine** — the moment a customer's stress score crosses a threshold, the recommendation engine is hard-blocked from pushing credit products, full stop.

This is also why we're keeping the three challenge points from the brief as three tightly-coupled engines sharing one customer feature store, rather than three independent tools — a stressed customer's chatbot tone should change, and a customer who's just had a life event should get a different recommendation ranking, automatically.

### 2.1 Who We're Designing For

| Persona                                           | Profile                                                              | Core pain point                                     | What moves the needle                                             |
|---------------------------------------------------|----------------------------------------------------------------------|-----------------------------------------------------|-------------------------------------------------------------------|
| <b>Ramesh — small farmer</b>                      | 45, UP, seasonal income, just moved from feature phone to smartphone | Confusing app, no formal credit history, fears debt | Crop-cycle aligned loans, Hindi voice support                     |
| <b>Priya — first-time salaried</b>                | 24, Tier-2 city, Rs. 18K/month, first job                            | No savings habit, low financial literacy            | Auto-save nudges, plain-language SIP suggestions                  |
| <b>Suresh — kirana store owner</b>                | 38, Tier-3 town, daily cash flow, GST filer                          | Needs working capital fast, hates paperwork         | Pre-approved OD from cash-flow, chat-based application            |
| <b>Meena — SHG member</b>                         | 35, rural Rajasthan, joint-liability group                           | No individual credit score, group dynamics          | Hybrid group + individual scoring, micro-insurance                |
| <b>Arjun — gig worker</b>                         | 28, delivery partner in a Tier-2 city                                | Irregular income, no PF/ESI safety net              | Income-smoothing micro-loans, flexi-EMI                           |
| <b>Lakshmi — elderly pensioner (our addition)</b> | 63, Tier-2 town, pension credit, low digital confidence              | Fears “pressing the wrong button”, avoids apps      | Voice-first flows, large-text mode, a human fallback one tap away |

*We added Lakshmi deliberately: pensioners and older users are a large, low-risk, high-trust-need segment that most fintech personalization case studies skip in favour of gig workers and first jobbers. If vernacular-first design only optimises for younger digital natives, it misses a chunk of exactly the population the brief is asking us to include.*

### 3. Solution Architecture

The architecture is one pipeline with five layers. Everything a customer sees in the output layer can be traced back through a specific model, a specific feature, and a specific consent record — that traceability is what makes the explainability and RBI-compliance criteria possible later, rather than bolted on at the end.

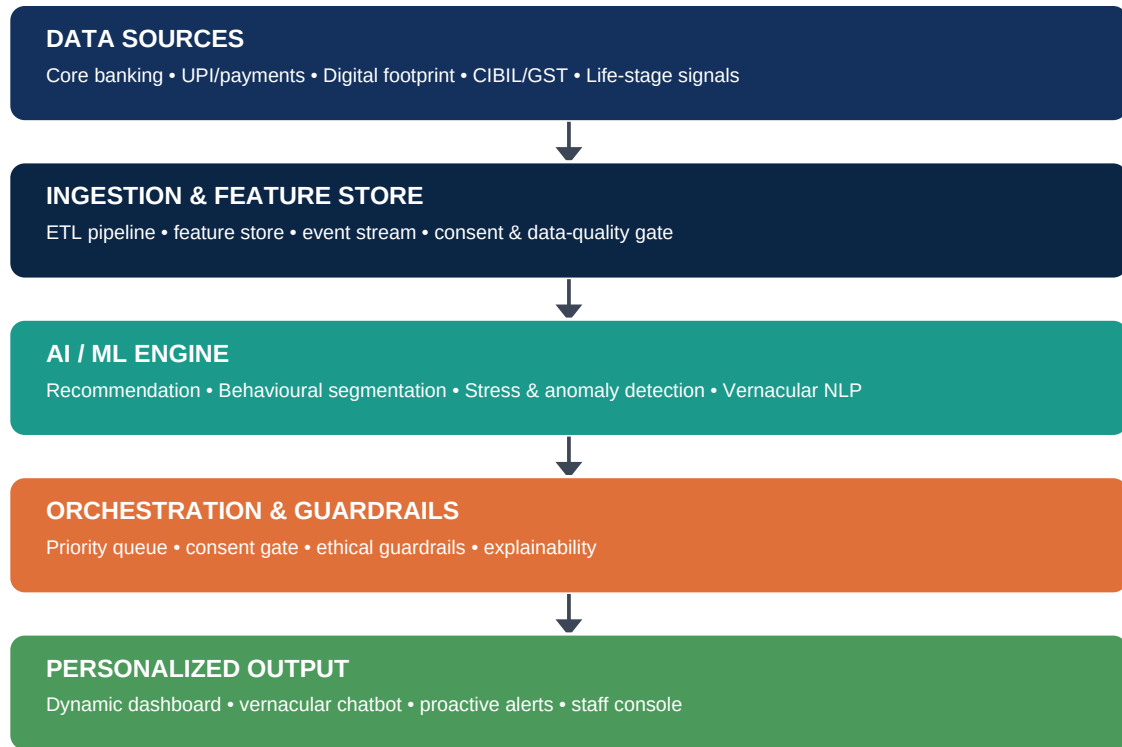


Fig. 1 — Data sources flow up through ingestion, the AI engine and a guardrail layer before reaching the customer or bank staff.

#### 3.1 Layer notes

| Layer                                 | What lives here                                                                                                                | Why it matters                                                                                             |
|---------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|
| <b>Data sources</b>                   | Core banking (txns, accounts, EMI), UPI/payments, app clickstream, external bureau/GST data, life-stage signals (SMS/UPI push) | Nothing is invented — the bank already owns almost all of this data today                                  |
| <b>Ingestion &amp; feature store</b>  | ETL pipeline, a shared feature store, an event stream for real-time triggers, and a consent/data-quality gate                  | One customer feature vector feeds all three AI engines — no duplicated logic, no drift between them        |
| <b>AI/ML engine</b>                   | Recommendation engine, behavioural segmentation, stress & anomaly detection, vernacular NLP                                    | The three challenge points, built as one coordinated system, not three silos                               |
| <b>Orchestration &amp; guardrails</b> | Priority queue (urgency × relevance × channel fit), consent gate, ethical guardrail checks, explainability generation          | This is where “personalization vs. genuine benefit” is actually enforced in code, not just in a policy PDF |
| <b>Personalized output</b>            | Dynamic dashboard, vernacular chatbot, proactive alerts, staff console for RMs/collections                                     | Same intelligence, different surface for the customer vs. the bank employee                                |

## 4. The Three Engines — Our AI/ML Approach

### 4.1 Engine 1 — Hyper-personalized product recommendations

The goal isn't “predict what this customer will click” — it's “recommend only what this customer can afford and is likely to actually need, at a moment when it's useful to know.” That reframing changes the model choice: affordability and moment-detection matter as much as click-through propensity.

#### Feature engineering

- RFM (recency / frequency / monetary) on spend behaviour
- Spend-category mix: food, health, education, transport, farm input
- Income stability — coefficient of variation of salary/UPI credits
- EMI burden ratio — total EMI ÷ monthly income
- Savings rate — monthly savings ÷ income
- Life-stage signal — marriage, new child, job change, retirement, inferred from spend patterns
- Digital maturity — app sessions, UPI usage frequency, bill-pay count
- Seasonality — festival spending spikes, crop cycle for farmers

#### Modelling choices

| Component                   | Approach                                                                                                               | Why this and not something fancier                                                                                      |
|-----------------------------|------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|
| <b>Behavioural segments</b> | K-Means (K chosen via silhouette score) with a Gaussian Mixture layer for soft/overlapping segments                    | Interpretable segments the product and compliance teams can actually explain to RBI, versus a black-box embedding space |
| <b>Candidate generation</b> | Content-based matching (TF-IDF/cosine on product attributes) + collaborative filtering (matrix factorisation)          | Handles both “customers like you also hold” and “this product literally fits your profile” cases                        |
| <b>Final ranking</b>        | A learning-to-rank model (gradient-boosted trees) combining relevance, affordability and propensity into one score     | Easy to attach SHAP-style feature attributions per recommendation for the explainability requirement                    |
| <b>Cold start</b>           | Rule-based life-stage + segment mapping until 3 months of transaction history exist, then graduated into the ML ranker | New-to-bank and new-to-digital customers are exactly who we can't afford to serve badly                                 |

#### Contextual trigger examples

- Salary credit + 2 days → “Start an SIP with just Rs. 500”
- Large one-off medical expense → health insurance suggestion for next time, not a loan upsell
- Balance sitting above 3× the customer's average → “Lock this surplus into an FD”
- EMI ending in 2 months → “Rs. X/month is about to free up — invest it or step it up?”

Every candidate recommendation is filtered before it ever reaches the ranker: already-owned products, active cool-down periods, missing consent, and — critically — the customer's current stress score are all checked first (Section 6 covers this gate in detail).

### 4.2 Engine 2 — Vernacular conversational AI for onboarding, loans and KYC

60–70% of loan applications get abandoned mid-journey, and the brief is right to connect this to language and complexity, not just UI polish. Our approach treats the chat interface as the primary journey for onboarding and loan applications — with the traditional multi-screen form as the fallback, not the other way around.

| Stage                        | Approach                                                                                                                                                             |
|------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Language detection</b>    | fastText language identification — fast, handles code-mixed text (“Hinglish” is the norm, not the exception)                                                         |
| <b>Voice input (ASR)</b>     | Whisper (or an Indic-tuned variant) — robust to Indian accents and code-switching                                                                                    |
| <b>Intent classification</b> | IndicBERT fine-tuned on banking intents across the priority languages: balance check, loan application, application tracking, fraud report, EMI query, general query |
| <b>Entity extraction</b>     | Fine-tuned multilingual NER for amounts, tenure, product names, dates                                                                                                |
| <b>Response generation</b>   | Template-based responses filled with customer context, with a tightly-guardrailed LLM fallback only for queries the templates can't cover                            |
| <b>Voice output (TTS)</b>    | Indic TTS for a natural-sounding voice reply on the voice channel                                                                                                    |

### Design rules we're holding the chatbot to

- No banking jargon — “EMI” becomes “mahina ka kitna bharna hai”, not a translated acronym
- Every number shown as digits *and* words, plus a visual slider where relevant
- Voice-first for low-literacy users, text as the alternative — not the default
- Every money-moving action is confirmed twice before it executes
- The bot always tells the customer which language it's replying in, and lets them switch mid-conversation

### 4.3 Engine 3 — Financial stress and fraud detection with empathetic intervention

This is the module we think matters most for the “genuine benefit” judging criterion, because it's the one place where the AI is explicitly asked to act against the bank's short-term revenue interest — catching stress before EMI bounce, and responding with support rather than a default flag.

#### Signals that feed the composite stress score

- **EMI stress** — EMI-to-income ratio above 40%, EMI bounces in the last 3 months, balance dipping just before the EMI date
- **Spending anomaly** — sudden ATM withdrawal spikes, new/unusual merchant categories, spend velocity at 2× the customer's average, late-night transaction clusters
- **Income disruption** — salary delayed more than 15 days, salary amount down more than 20%, no credits for 60+ days for an otherwise-regular salaried customer
- **Behavioural shift** — app login frequency change, a new spending geography, multiple loan inquiries in a short window (credit-hungry signal)

#### Modelling choices

| Component                 | Approach                                                                                                                                                 |
|---------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Point anomalies</b>    | Isolation Forest on transaction-level features — fast, no distribution assumptions, works well on the tabular banking features we already have           |
| <b>Sequence anomalies</b> | An LSTM autoencoder over 30-day transaction windows; a high reconstruction error flags a behaviour pattern that's broken from the customer's own history |
| <b>Slow deterioration</b> | CUSUM / EWMA on daily balance and spend — catches a gradual downward drift that a point-anomaly model would miss entirely                                |
| <b>Composite score</b>    | A weighted index (0–100) combining all four signal families, with weights tuned against historical default/restructure outcomes                          |

#### The intervention ladder



Fig. 2 — Higher stress scores escalate to more human, less automated intervention — never to more selling.

Crucially, crossing into Yellow doesn't just add a banner — it flips a switch in Engine 1 that blocks new credit-product recommendations for that customer until the score drops back down. That cross-engine wiring is the whole point of sharing one feature store.

## 5. What We're Adding Beyond the Brief

*The three engines above answer the challenge as written. The ideas in this section are ours, on top of it — things we think earn real points on “innovation” and “genuine benefit” without diluting focus on the core three problems.*

### 5.1 A customer-facing “why am I seeing this” explainer

Most explainability work in these solutions (including in our own architecture above) is aimed at bank staff and regulators — a SHAP chart an RM or auditor can read. We want to push explainability one layer further, to the customer, in their own language: every recommendation and every stress alert carries a one-line, plain-language reason — “Hum yeh sujhaav de rahe hain kyunki pichle 3 mahine mein aapki income badhi hai” (“We’re suggesting this because your income has grown over the last 3 months”) — generated directly from the top feature attribution, not a canned line. This directly targets the trust deficit named in the brief’s background section, and it’s a cheap addition once SHAP values already exist for the staff dashboard.

### 5.2 A visible “personalization dial”

Alongside DPDP consent toggles (which are necessary but usually buried in settings), we’d add one always-visible control on the dashboard: a slider from “show me less” to “show me more” personalized suggestions, plus a one-tap “mute this category for 90 days” on any recommendation card. It’s a small UI idea, but it converts consent from a legal checkbox into something the customer actually feels in control of — which is what “not just upsell-driven” should look like from the customer’s seat, not just the bank’s compliance seat.

### 5.3 Community-assisted onboarding (“Digital Sakhi” channel)

For the segments most likely to abandon a digital journey — Ramesh and Lakshmi in our persona set — we’d route a portion of onboarding through existing bank correspondents / SHG animators, giving them a simplified staff-side version of the same chatbot to co-pilot the customer’s onboarding in person. This keeps a human in the loop exactly where trust is weakest, while all the same AI personalization still runs underneath — it’s a channel decision, not a separate system.

### 5.4 Other additions carried through from the original brainstorm

- **Family Financial Health Score** — an opt-in, consent-gated household view (emergency-fund coverage, insurance adequacy, debt health, savings rate, goal progress) shown as a single 0–100 score with a clear “how to improve it” path, gamified lightly with progress badges.
- **WhatsApp / USSD channel** — balance checks, recommendations and loan applications over WhatsApp Business API, with a \*99# USSD fallback and a missed-call-for-mini-statement option for feature-phone users.
- **SHG / community intelligence** — group repayment-discipline tracking, flagging which SHG members may need support, and recommending group vs. individual insurance.
- **60-second financial literacy micro-lessons** — vernacular, contextually triggered (e.g. right after a missed EMI: “how to manage EMI stress”), with a short quiz and reward points.
- **Predictive life-event engine** — jewellery + banquet-hall spend → likely wedding; hospital + paediatrician spend → likely new child; GST + new-supplier payments → growing business — each mapped to a relevant, non-intrusive product suggestion.

## 6. Ethical Safeguards — Data Privacy, Bias and Anti-Predatory Design

We're treating this as a hard gate every recommendation and alert must pass through before it reaches a customer — not a policy document that sits next to the code. Concretely, the gate enforces:

| Safeguard                                | What it enforces                                                                                                                                                                                        |
|------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Consent &amp; DPDP Act compliance</b> | Granular, opt-in consent per data category (transaction data, AI recommendations, stress monitoring, vernacular comms); every consent has an expiry and a one-tap revoke; no pre-checked boxes anywhere |
| <b>RBI data localization</b>             | All customer data and model artefacts stored and processed in-region (India); the gate rejects any pipeline step that would move data outside that boundary                                             |
| <b>Stress-lock on credit products</b>    | Any customer whose stress score is above the Yellow threshold cannot be shown a <i>new</i> loan/credit-card recommendation — only restructuring, refinancing or counselling options                     |
| <b>Cool-down periods</b>                 | A rejected or ignored recommendation isn't repeated for at least 30 days, and no more than 3 recommendations are shown in a single session                                                              |
| <b>No dark patterns</b>                  | Automated language checks reject urgency wording ("expires in 10 minutes", "last chance") and any pre-checked consent state before a message ships                                                      |
| <b>Explainability by default</b>         | No recommendation or stress alert reaches a customer or an RM without an attached, feature-level explanation — unexplainable output is blocked, not just discouraged                                    |
| <b>Algorithmic fairness audits</b>       | Model accuracy is checked separately across Tier 1/2/3/rural segments before deployment; a segment falling below a fairness threshold blocks release until addressed                                    |
| <b>Full audit trail</b>                  | Every AI decision — recommendation shown, alert raised, consent change — is logged with model version and consent status, for RBI examination on demand                                                 |

The stress-lock and the no-dark-patterns check are the two we'd defend hardest in front of judges: they are the concrete answer to "not just upsell-driven," because they describe situations where the AI is *not allowed* to make the bank money, by design.

## 7. Prototype / Wireframe Plan

For the working-prototype deliverable, we'd rather demo four tight customer-journey screens end to end than a wide, half-finished app. In priority order:

| Screen / flow                                    | What it shows                                                                                                                                             |
|--------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>1. Segment-aware dashboard</b>                | Widget order and content changes with the customer's segment — a stress alert banner takes the top slot the moment it exists, ahead of any recommendation |
| <b>2. Vernacular chatbot</b>                     | A Hindi (plus one more language) conversation handling balance check, loan query and EMI calculator intents, with a visible language switcher             |
| <b>3. Conversational loan journey</b>            | Amount → tenure → EMI confirmation → KYC, as a guided step flow with a visual EMI slider instead of a static form                                         |
| <b>4. Stress alert &amp; restructuring offer</b> | The Yellow/Orange intervention screen — empathetic copy, an EMI-date-shift option, and a visible “why am I seeing this” explanation                       |

Supporting artefacts alongside the four screens: the architecture diagram from Section 3, the intervention-ladder diagram from Section 4.3, and a short synthetic dataset (a few thousand generated customers and transactions) so the recommendation and stress models have something real to run against in the demo, since production banking data obviously isn't available for a hackathon.

## 8. Tech Stack Snapshot

| Layer                | Choices                                                                                                                                                                                  |
|----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Frontend</b>      | React + Vite, TailwindCSS, i18next for the vernacular strings, Recharts for the spending insights, a PWA service worker for offline/low-bandwidth mode                                   |
| <b>Backend</b>       | Django + DRF for REST APIs, Django Channels for the chat WebSocket, Celery + Redis for async scoring jobs                                                                                |
| <b>Data</b>          | MongoDB Atlas — customers, transactions, recommendations, stress_alerts, consent_logs and audit_trail as separate collections                                                            |
| <b>ML / training</b> | scikit-learn, XGBoost/LightGBM, TensorFlow (for the LSTM autoencoder), HuggingFace Transformers (IndicBERT), SHAP for explanations, trained in Colab and shipped as serialized artefacts |
| <b>Deployment</b>    | Frontend on Vercel, backend on Render/Railway, models loaded from saved artefacts at API start-up, GitHub Actions for CI/CD                                                              |

## 9. Mapping Back to the Judging Criteria

| Criterion                                                   | Where this document answers it                                                                                                                                                                                                                                                                      |
|-------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Innovation &amp; technical feasibility</b>               | Three coupled engines on one shared feature store (Sec. 3–4), plus the customer-facing explainer, personalization dial and Digital Sakhi channel as original additions (Sec. 5); every model choice is a standard, production-proven algorithm — nothing here needs a research breakthrough to ship |
| <b>Personalization depth vs. genuine benefit</b>            | The stress-lock on credit recommendations (Sec. 4.3, 6) and the cool-down/session limits are concrete mechanisms, not intentions — the system is designed to sometimes recommend nothing                                                                                                            |
| <b>Explainability &amp; RBI compliance readiness</b>        | Feature-level explanations required before any output ships, a full audit trail, data localization and DPDP-compliant granular consent (Sec. 6)                                                                                                                                                     |
| <b>Usability for non-tech-savvy, vernacular-first users</b> | Vernacular-by-default chatbot design rules (Sec. 4.2), the Lakshmi persona and Digital Sakhi channel (Sec. 2.1, 5.3), WhatsApp/USSD fallback (Sec. 5.4)                                                                                                                                             |
| <b>Scalability &amp; social/financial impact</b>            | Managed, auto-scaling data layer, async job processing, model versioning (Sec. 8); impact case is early stress intervention preventing default cascades and vernacular access closing the digital divide (Sec. 2, 6)                                                                                |

## 10. Closing Note

The brief's background section makes a sharp observation: banks already have the data, they just serve everyone the same experience anyway. Our bet is that the winning entry here won't be the one with the fanciest model — recommendation engines, anomaly detectors and Indic NLP are all well-trodden ground individually. It'll be the one that can show, concretely, moments where the AI chose *not* to sell something, and where a customer in a language other than English could get through a loan application without giving up halfway. That's the version of this project we want to build.

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— Team Hack Horizon